

Low-Cost Deployable Seafloor Metal Detection Sensor

A Pulse Induction Based Platform for Ocean Resource Exploration

Technical Project Report

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Team Name: BlueEcho

Team ID: 182761

College: Dr B R Ambedkar National Institute of Technology Jalandhar

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Abstract

This report details the design and implementation of a low-cost, real-time Seafloor Magnetic Anomaly Detection System prototype. The architecture utilizes a custom Arduino-based Pulse Induction (PI) metal detector, a Python-based serial data bridge with real-time signal processing, and a dynamic Streamlit dashboard for data visualization and anomaly tracking. The system is designed to demonstrate the feasibility of detecting magnetic anomalies in marine environments while dynamically adapting to baseline drift and mitigating electromagnetic interference (EMI).

1 Introduction

Marine exploration and underwater target localization require robust, interference-resistant magnetic detection systems. Traditional solutions are often cost-prohibitive or lack real-time adaptability.

This project proposes a hardware-software co-designed system that pairs a Pulse Induction (PI) sensor with intelligent software baseline calibration, allowing for sensitive detection with a noise-reduced baseline suitable for real-time monitoring and data logging.

2 Hardware Architecture

The physical sensor relies on the principle of Pulse Induction. A magnetic field is generated using a pulsed transmitter coil. When a conductive metallic object is nearby, eddy currents are produced inside the target. These currents generate a secondary magnetic field that influences the measured decay response.

2.1 Analog Front-End Circuitry

The hardware prototype consists of the following major stages:

- **Search Coil:** A custom-wound copper coil is used to generate and detect changes in the magnetic field.
- **Switching Stage:** An IRF520 MOSFET is used to rapidly switch current through the transmitter coil, producing the excitation magnetic pulse.
- **Damping and Protection:** A damping resistor is used to reduce unwanted ringing after the MOSFET switches off. Clamping diodes protect the receiver circuitry from voltage spikes generated during coil discharge.
- **Amplification:** An LM324 operational amplifier forms the analog receiver front-end. It amplifies the received signal before it is sampled by the Arduino's Analog-to-Digital Converter (ADC).

2.2 Microcontroller Pulse Timings

Precise timing is used to separate the excitation pulse from the subsequent magnetic decay signal:

- **Pulse Width:** 120 μs

- **Decay Delay:** $20\ \mu\text{s}$
- **Sampling Cycle:** Approximately 20 ms between measurement cycles

The MOSFET remains ON during the pulse interval and is switched OFF before the ADC reading is taken. The short delay between switching and sampling allows the large initial transient to reduce before measurement.

3 Data Pipeline and Signal Processing

Raw analog readings from the PI circuit may contain high-frequency fluctuations, baseline drift, and electromagnetic interference.

A digital signal-processing pipeline is therefore used to stabilize the signal before visualization and anomaly detection.

3.1 Firmware EMA Filter

At the microcontroller level, a software Exponential Moving Average (EMA) filter is applied to smooth the ADC readings before serial transmission.

The recursive filtering equation is:

$$S_t = (0.6 \times S_{t-1}) + (0.4 \times R_t) \quad (1)$$

where:

- S_t is the current smoothed signal,
- S_{t-1} is the previous smoothed signal,
- R_t is the latest raw ADC reading.

The weighting provides a balance between signal stability and response speed. This helps suppress high-frequency fluctuations while preserving changes produced by nearby metallic objects.

3.2 Python Serial Bridge

The filtered serial data is received by a Python-based serial bridge.

The bridge performs three primary functions:

- receives serial data from the microcontroller,
- stores measurement data for later analysis,
- forwards the data to the Streamlit visualization interface.

Instead of relying only on a fixed absolute threshold, the system forwards the measured values to the dashboard, where an adaptive baseline is used for anomaly detection.

4 Streamlit Visualization Dashboard

The visualization interface is implemented using Streamlit.

It provides real-time signal plotting, baseline calibration, anomaly detection, and continuous monitoring of the detector output.

4.1 Dynamic Baseline Calibration

Environmental conditions and electronic characteristics may cause the ADC baseline to vary over time.

To compensate for this variation, the dashboard performs a short calibration period.

During calibration, multiple signal samples are collected over a five-second interval and used to calculate the baseline:

$$\text{Baseline} = \frac{1}{N} \sum_{i=1}^N \text{Signal}_i \quad (2)$$

where N represents the number of samples collected during calibration.

The calculated value becomes the reference level for subsequent anomaly detection.

4.2 Hysteresis-Based Anomaly Detection

A hysteresis mechanism is used to reduce repeated detection of the same target when the measured signal fluctuates around the threshold.

The current thresholds are:

- **Trigger Threshold:**

$$\text{Baseline} + 10$$

- **Reset Threshold:**

$$\text{Baseline} + 2$$

An anomaly event is generated when the signal rises above the trigger threshold.

After detection, the system remains in the anomaly state until the signal falls below the reset threshold.

This separation between the two threshold levels reduces rapid switching between normal and anomaly states.

5 Overall System Workflow

The complete system operates through the following sequence:

1. The Arduino generates a controlled excitation pulse.
2. The IRF520 MOSFET switches current through the transmitter coil.

3. The coil produces a magnetic field.
4. A nearby conductive metallic object develops induced eddy currents.
5. These eddy currents produce a secondary magnetic response.
6. The receiver circuitry captures the resulting decay signal.
7. The LM324 amplifies the received response.
8. The Arduino samples the amplified voltage using its ADC.
9. An EMA filter reduces high-frequency fluctuations.
10. The processed values are transmitted through the serial interface.
11. A Python program receives and logs the incoming data.
12. Streamlit calculates the local baseline.
13. Hysteresis-based logic identifies significant signal deviations.
14. Detected anomalies are displayed to the user in real time.

6 Conclusion

The proposed system demonstrates a low-cost hardware-software approach to Pulse Induction based anomaly detection.

The combination of custom sensing hardware, controlled coil excitation, analog amplification, microcontroller-based signal filtering, adaptive baseline calibration, and real-time Streamlit visualization provides an integrated platform for experimental marine sensing applications.

Rather than relying exclusively on a fixed detection threshold, the system uses a locally measured baseline as its reference. This allows the detection logic to adapt when the measured signal changes because of electronic drift, environmental conditions, or electromagnetic interference.

The prototype demonstrates the feasibility of combining inexpensive electronics with software-based signal processing to create an accessible platform for metallic anomaly monitoring and further development toward marine exploration applications.

References

- [1] “Design and Analysis of Pulse Induction Underground Mines Detection System,” *International Conference on Functional Materials and Metallurgy*, Pulau Pinang, September 2012.
- [2] “Underwater Multi-Parameter Magnetic Anomaly Detection System Carried by Autonomous Underwater Vehicle and Its Data Preprocessing Method.”